

Vatsal Yadav

linkedin.com/in/vatsal-yadav | github.com/vatsalyd | vatsalyd.github.io/Portfolio
LeetCode: leetcode.com/u/vatsal_yd | Medium: medium.com/@vatsal.y.official

Email: vatsal.y.official@gmail.com
Mobile: +91-79837-09173

PROFESSIONAL SUMMARY

IIT Bhilai B.Tech student focused on **AI/ML, LLM applications, and AI systems**, with hands-on experience building **RAG pipelines, multi-agent systems, multimodal AI, model-serving APIs, and ML models**; targeting **AI/ML and LLM engineering intern roles**. Strong in Python, PyTorch, LangGraph, FastAPI, vector search, Docker, and AWS, with production internship experience and measurable system improvements.

EDUCATION

Indian Institute of Technology (IIT) Bhilai Bhilai, Chhattisgarh
Bachelor of Technology in Data Science & Artificial Intelligence; GPA: 7.34/10 2024 – 2028
Coursework: Data Structures, Algorithms, Computer Organisation & Architecture, Statistical Programming, Data Analytics & Visualization

EXPERIENCE

Incrivelsot Private Limited (NUMAA.ai) Remote
AI & ML Intern May 2026 – July 2026
– Architected the **Nutritionist Lite Agent** as a 3-layer hybrid system: deterministic clinical engine (ICMR-NIN 2020 / WHO guidelines) → keyword-based guideline retrieval → **Google Gemini (gemini-2.5-flash)** orchestration router; enforced hard medical-nutrition boundaries to reduce LLM hallucinations in production recommendations.
– Engineered 5 production services (**MealPlanningService, ChatService, VisionService, NutritionUIService, NutritionEngine**) with **FastAPI + Pydantic, MongoDB, ChromaDB** prototyping, **Qdrant RAG** in production, and a Gemini Vision multimodal endpoint with confidence-based fallback.
– Led migration from 3 LLM providers (Groq, Google, NVIDIA) to **Google Gemini**; implemented token-bucket rate limiting (300 RPM), circuit breaker (5 failures / 30s cooldown), and model fallback logic, eliminating HTTP 429 errors across production task routes.
– Applied **OOP** to refactor **NutritionUIService** (855 → 480 lines); added **31 unit tests** and a **FakeProfileRepository** adapter for MongoDB-free testing.

PROJECTS

Context Pager | AI/ML Engineer GitHub | Live
Python, MCP, LLMingua-2, BGE-M3, FastAPI, Docker, AWS
– Built an **MCP (Model Context Protocol)**-native document intelligence server reducing AI-agent token usage by **4–10x** via on-demand compressed paging; exposed document search, compression, and long-term memory tools for Claude Desktop, Claude Code, and Cursor.
– Architected a privacy-first relay on **AWS t3.micro**: local AI models → encrypted cloud routing → agent; masked PII via **Microsoft Presidio**, SHA-256 hashed API keys, and rate-limited at 100 calls/hour.

HelixDesk – Enterprise Support Intelligence | AI/ML Engineer GitHub | Live
Python, LangGraph, FastAPI, Pinecone, Docker, Llama-3.3-70b
– Designed and shipped a production-grade **RESTful API backend** using async FastAPI + Pydantic and a **3-agent LangGraph state machine** (Triage → Retrieval → Resolution); classified tickets across 4 categories and escalated cases below **50%** confidence.
– Implemented semantic vector search with **Pinecone** and multilingual-e5-large embeddings, citation-backed generation, structured JSON logging, **X-Correlation-ID** propagation, and **GitHub Actions CI/CD**; achieved 1840 ms average resolution.

JobFit-AI – Resume Matching Engine | ML Engineer GitHub | Live
Python, XGBoost, PyTorch, SBERT, Streamlit, Docker, AWS EC2
– Engineered a **3-model** resume-JD matching pipeline: rule-based skill scorer (**430+ skill dictionary**, 24 categories), **XGBoost** on 10 features, and fine-tuned **SBERT dual-encoder** based on **all-MiniLM-L6-v2**, using BatchNorm, GELU, dropout, OneCycleLR, and early stopping; trained on **13,000+ resume-JD pairs**.
– Applied rank-based label scoring (mean of 50, standard deviation of 21.5) for balanced calibration; containerized CPU-only PyTorch and deployed on **AWS EC2 (t3.small)** with Docker and auto-restart; served interactive inference via **Streamlit**.

TECHNICAL SKILLS

Languages: Python, C, JavaScript, SQL
AI/LLMs: LangChain, LangGraph, Multi-Agent Systems, RAG, MCP, Prompt Engineering, Fine-tuning, LoRA, PEFT, Multimodal AI, Vision-Language Models, Embedding Models, Google Gemini API, OpenAI API, Hugging Face
ML/Deep Learning: PyTorch, Scikit-learn, XGBoost, Transformer Architecture, Neural Networks, Sentence-Transformers (SBERT), NumPy, Pandas, NLP, EDA, Feature Engineering
Backend/DevOps: FastAPI, RESTful APIs, Pydantic, OOP, Docker, AWS (EC2, ECR), GitHub Actions (CI/CD), MongoDB, Qdrant, ChromaDB, Pinecone, Streamlit

LEADERSHIP & ACTIVITIES

Data Science and AI Club (DSAI), IIT Bhilai | Coordinator & Core Member Aug 2024 – Present
– Created and maintain an open-source **AI/ML Compendium** covering agentic AI, LLMs, and ML fundamentals; conducted technical sessions on **Agentic AI** and multi-agent systems; organized a **100+ participant** hackathon at Meraz, IIT Bhilai.

ACHIEVEMENTS

Campus Finalist – The Integral Cup (2024): Selected among top performers nationally in an algorithmic and quantitative reasoning competition by **Optiver, QRT, and Jane Street**.